



Considerations about wind gust thresholds related to social impact: study of different regions in Catalonia

Barbería L. (1), Amaro J. (1), Aran M. (1) and Llasat M.C. (2)

(1) Meteorological Service of Catalonia, Barcelona. Spain; (2) Dept. Astronomy & Meteorology. Univ. Barcelona. Spain

Correspondence to: L. Barbería (lbarberia@meteo.cat)

1. INTRODUCTION

Severe weather can cause several damages on a territory and its population (Munich Re, 2016)
WIND is one of the most important meteorological phenomena which cause remarkable economic losses (Consorcio de Compensación de Seguros, 2015)

PREVIOUS STUDIES

Drawback:
vulnerability due to population was not taken into account

- Fixed wind thresholds:** MEDEX (Jansa et al, 2014), Catalan Meteorological Warning Alarms (Vilaclara et al, 2010)
- Different wind thresholds for different regions:** 98th percentile of daily maximum gust speed (Hewston et al, 2011), return periods (Stepak et al, 2012)

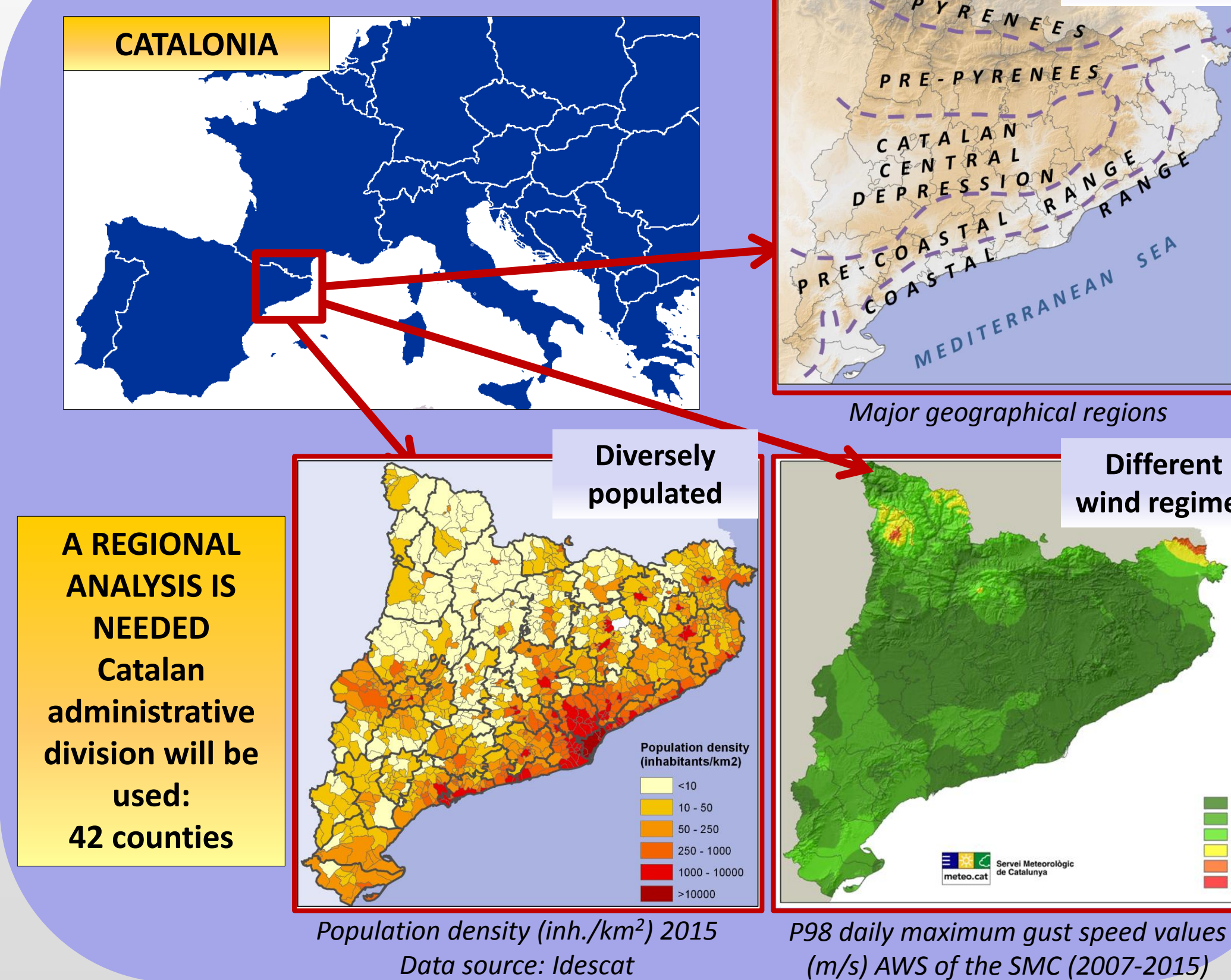
CURRENT STUDY

The vulnerability factor is included

AIM: To obtain, for each county in Catalonia, thresholds of gust speed from which a remarkable social impact is observed.

APPLICATIONS: Better knowledge of the gust speed values that cause damage in different areas in Catalonia. In the future it could be used to consider new thresholds for Civil Protection alarms. Therefore, a higher accuracy by region will be reached.

2. AREA OF STUDY



3. DATA AND METHODOLOGY

Studies conducted by the Social Impact Research Group, in the frame of MEDEX, stated that **requests related to damage claims** received in Meteorological Services are a **good proxy indicator of social impact** (Amaro et al, 2010)

➤ **Source:** Database of requests received in the Meteorological Service of Catalonia (SMC) 2011-2015

SMC GENERAL PROCEDURE

REQUESTS
Citizens, companies or organizations ask for data for a period in a certain location

EVENT ANALYSIS
Association of the most representative weather station of the SMC network for each event. Individual treatment of each request: **HIGHER ACCURACY**

METEOROLOGICAL REPORT
Report with the most representative data for the location requested

TOTAL NUMBER OF REQUESTS (2011-2015): 17.674

Criteria for the study database:

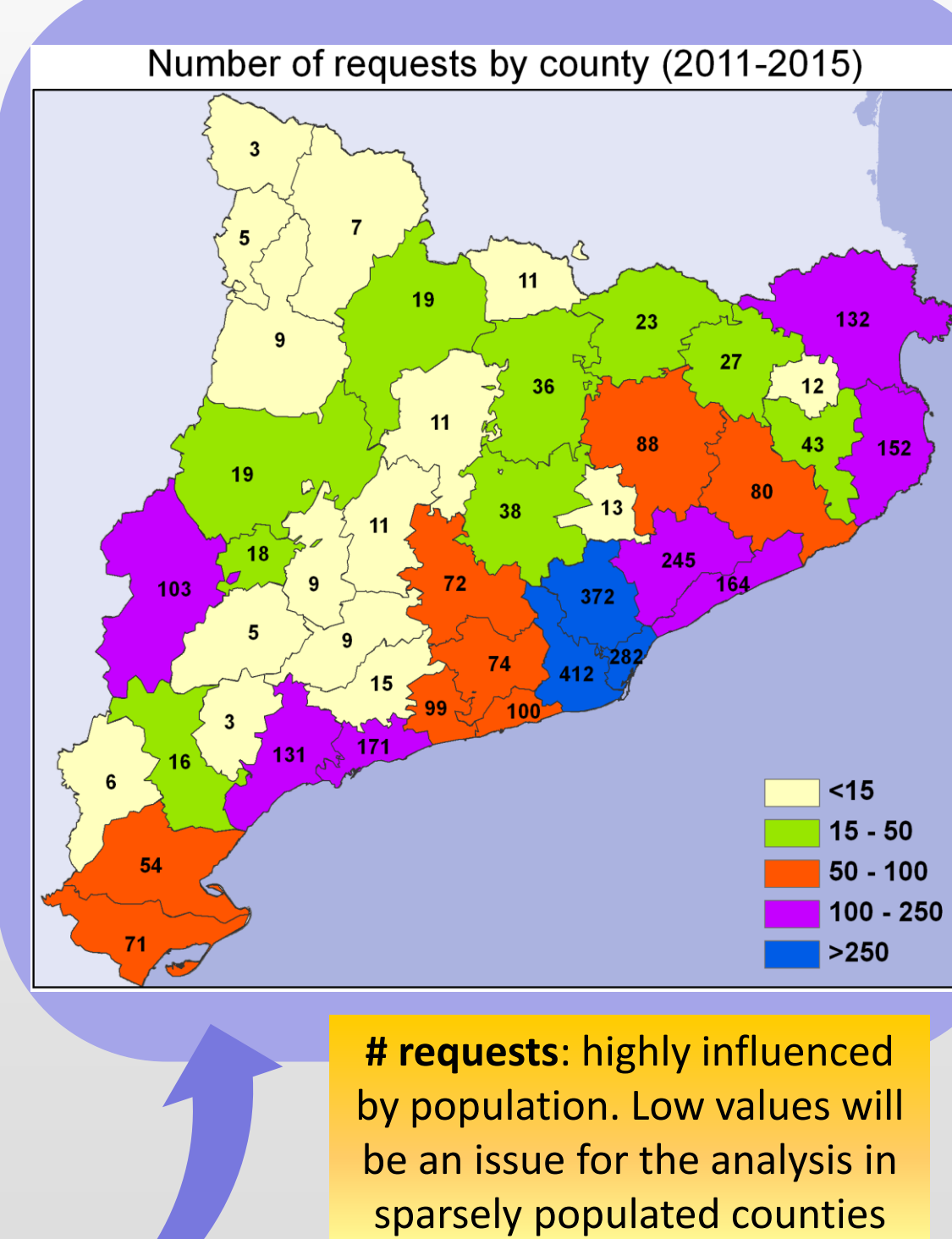
- Requests received between 2011 and 2015, related to events of any year
- Requests which only demand gust speed data
- Period requested: less than a month
- Not included if it was not related to a damage claim

CRITERIA

NUMBER OF REQUESTS CONSIDERED: 3.170

NEW DATABASE: REQUESTS vs GUST SPEED by county

Statistical analysis



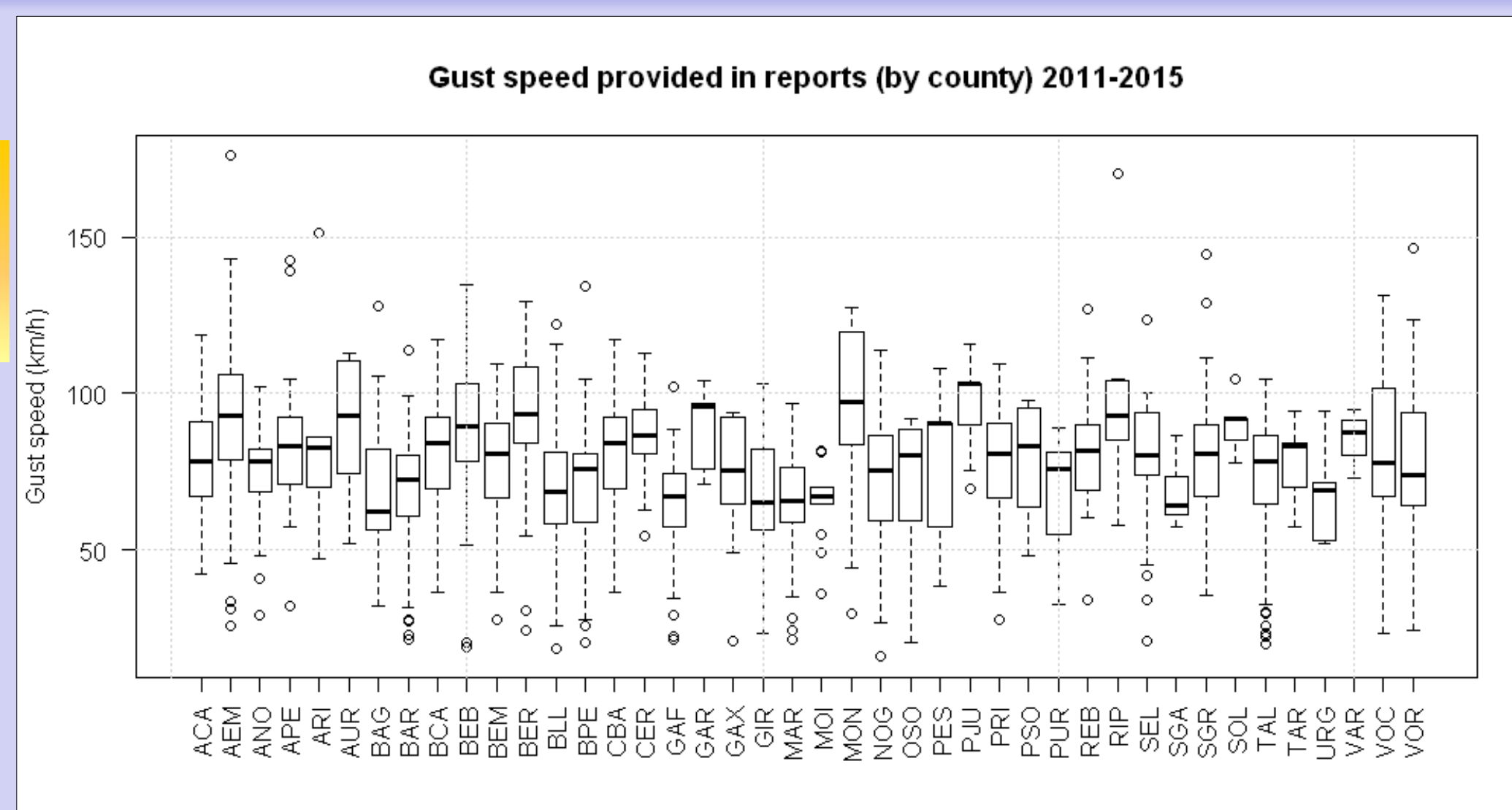
4. RESULTS: REQUESTS VS GUST SPEED

a) RANGE OF GUST SPEED VALUES

SUMMARY

- **Median: 76.1 km/h**
- **Mean: 77.2 km/h**
- **Standard deviation: 21.7 km/h**

- **Most of the gusts: 60-100 km/h (68% of the requests)**



Remarkable differences between counties

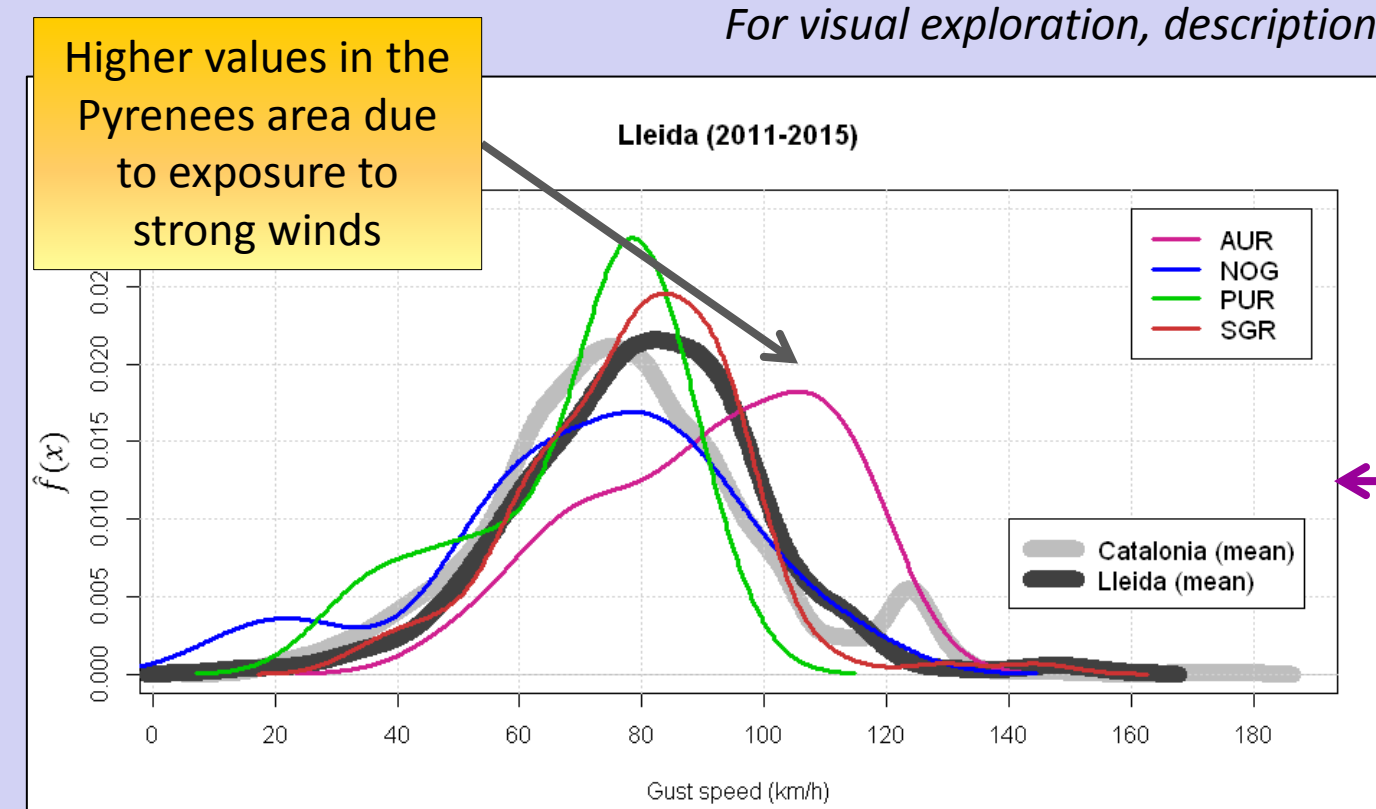
Confirm that a regional analysis is needed
Different thresholds should be defined

Example

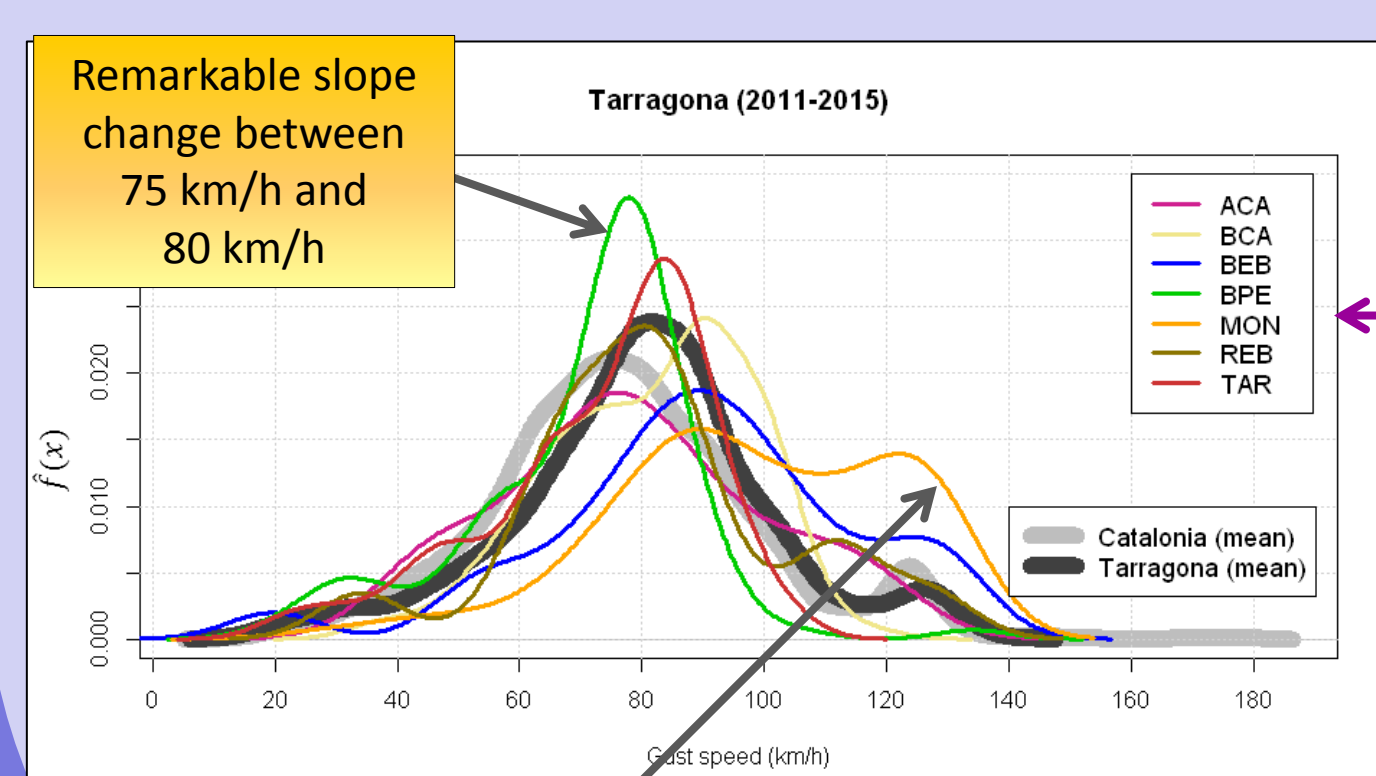
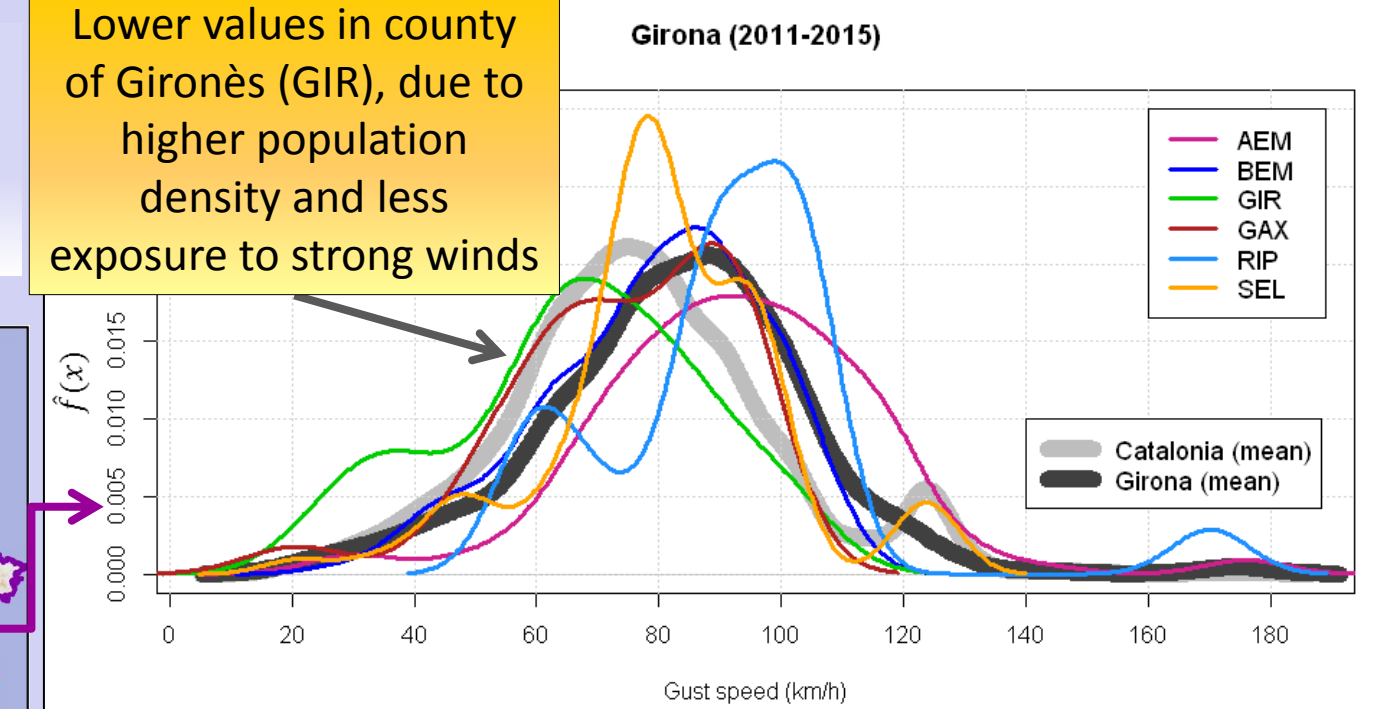
Median in Montsià (BAR) county: 72.4 km/h
Median in Montsià (MON) county: 97.4 km/h

b) KERNEL DENSITY ESTIMATION BY COUNTY (GROUPED BY PROVINCE)

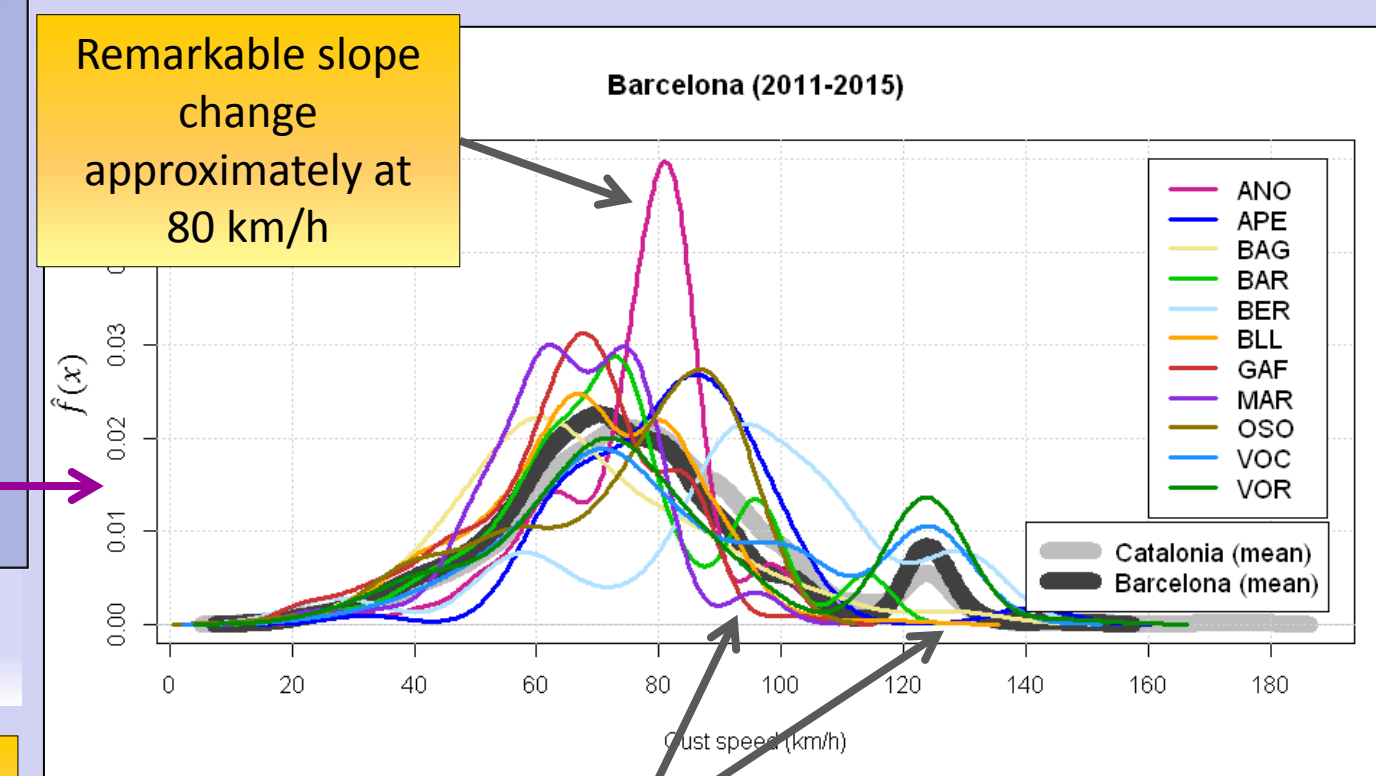
For visual exploration, description and comparison of the data (Gaussian Kernel smoothing, default bandwidth using R package)



Only calculated for counties with 15 or more requests
Comparison with mean values for Catalonia (grey lines) and for each province (black lines)



In violet, province administrative division: Lleida, Tarragona, Girona and Barcelona



High values in the extreme southern counties, highly exposed to strong winds

In general, a remarkable rise is observed in the interval 60-70 km/h

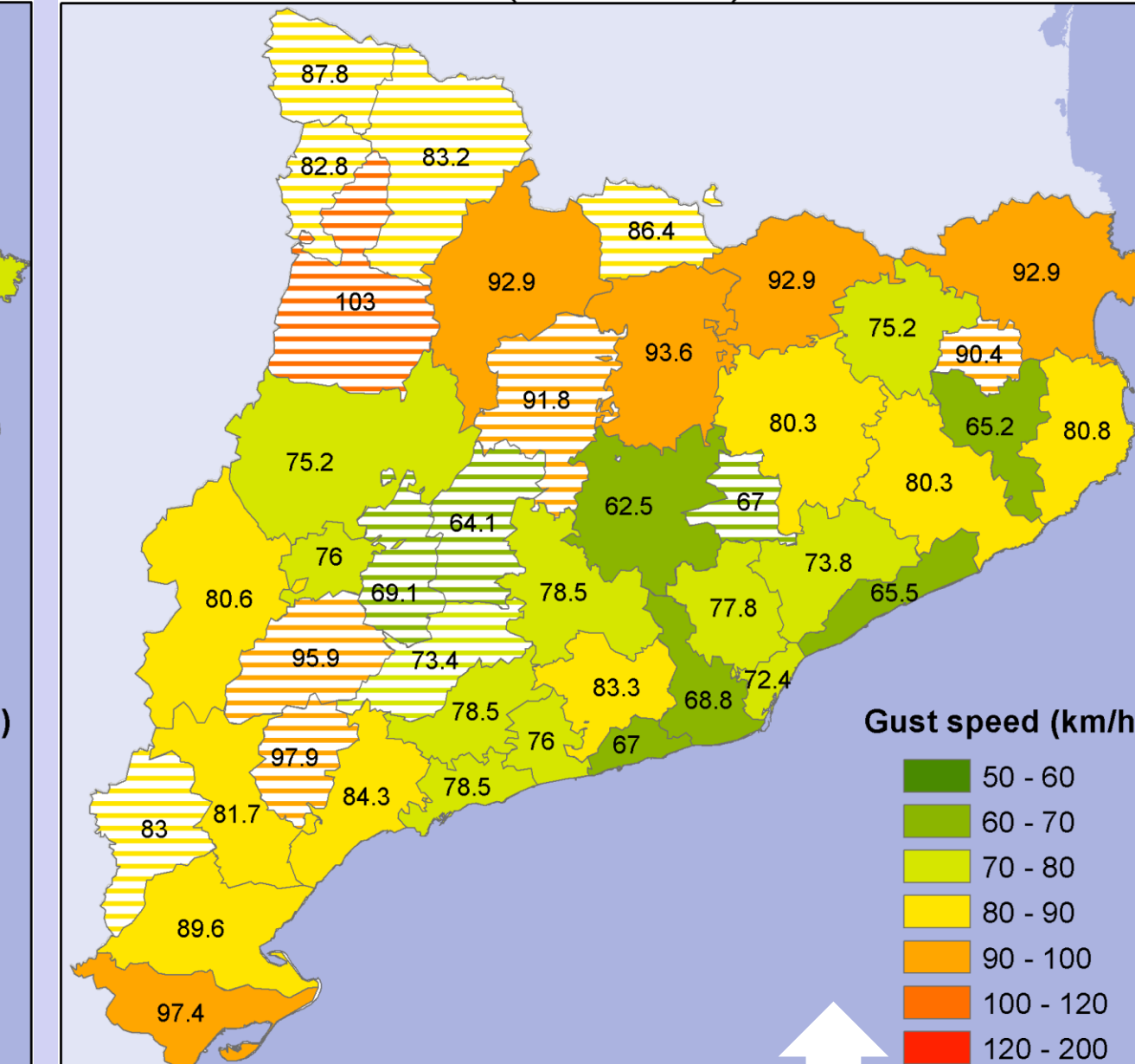
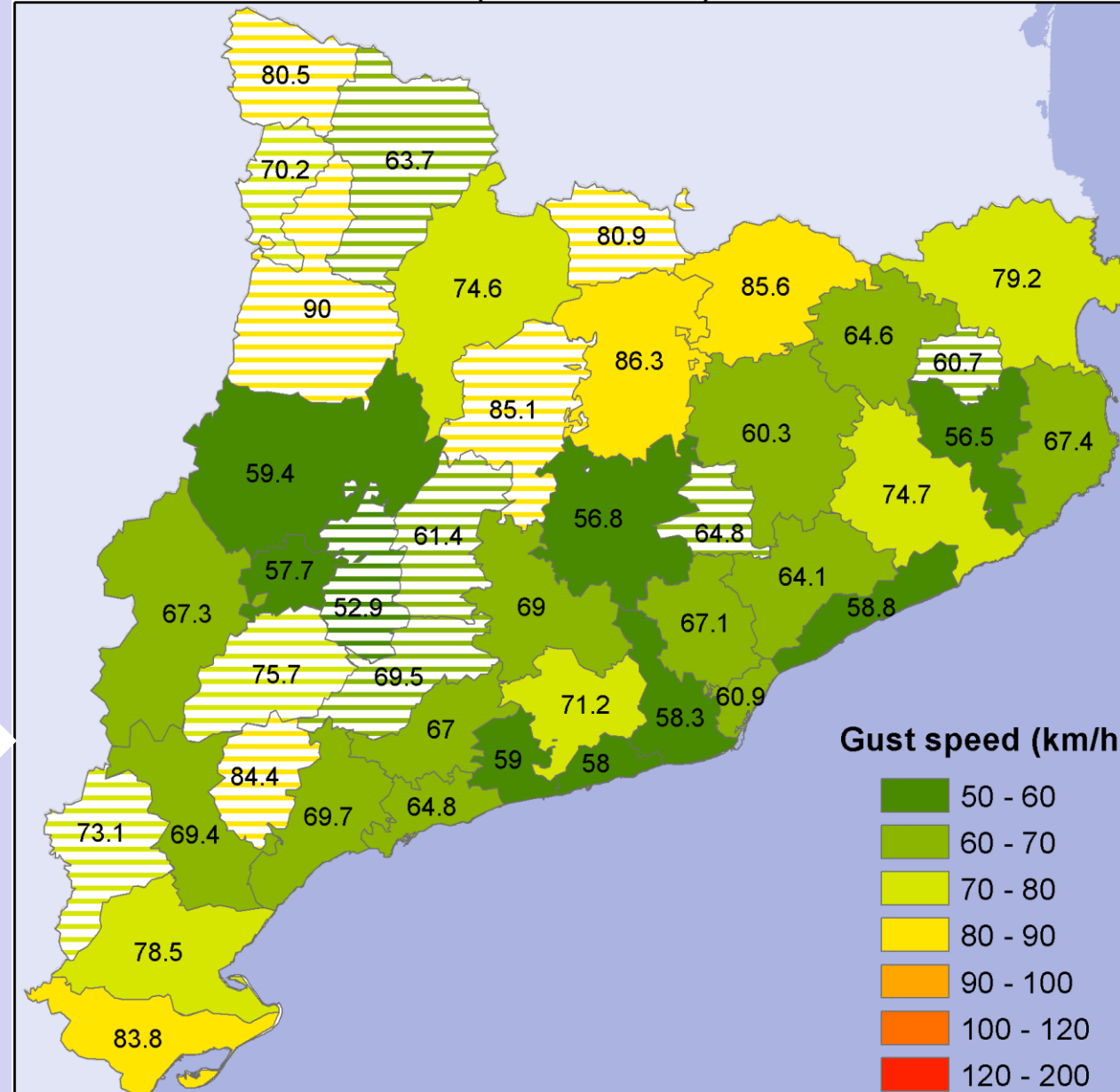
Most of the counties of Barcelona province show lower values than the rest of counties in Catalonia

Secondary peaks due to certain extreme events (especially in highly populated areas) and to different response inside a county

c) QUARTILE MAPS BY COUNTY

P25 Gust speed provided in reports (2011-2015)

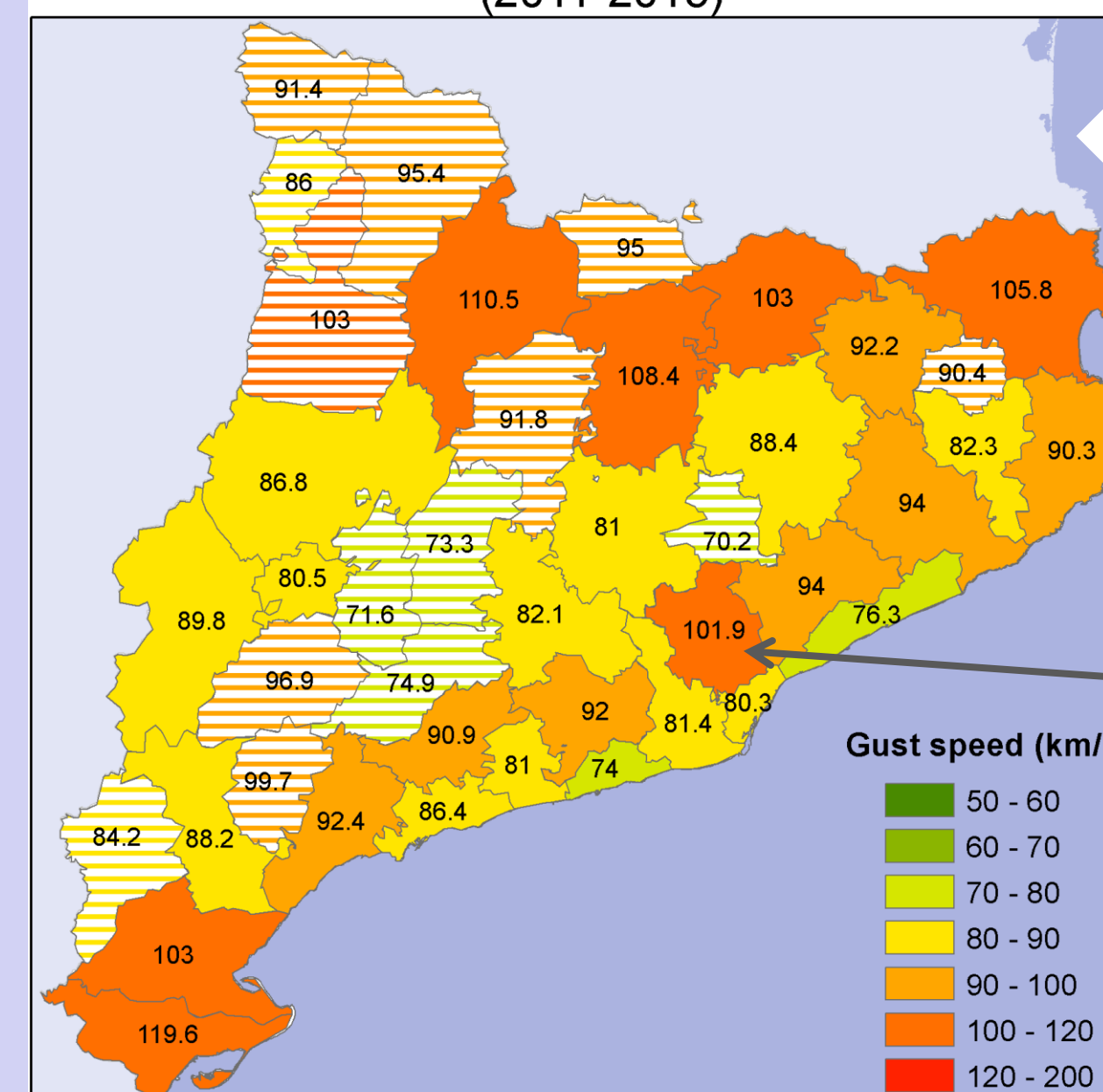
P50 Gust speed provided in reports (2011-2015)



SUMMARY OF APPROXIMATE VALUES

Area	P25 (km/h)	P50 (km/h)	P75 (km/h)
Pyrenees area and extreme northern and southern coastal counties	75-85	85-100	100-120
Central coastal area, county of Gironès and some counties in the Catalan Central Depression	55-65	65-75	75-85
Other areas	65-75	75-85	85-95

P75 Gust speed provided in reports (2011-2015)



Stripe pattern in counties with less than 15 reports

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